Manuela A. Pop

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SUMMARY

Experienced software engineer building AI-enabled systems, including retrieval-augmented generation, agentic workflows, LLM tool integration, ML pipelines, and optimization systems. Combines 15+ years of platform engineering leadership with current graduate work in artificial intelligence to deliver scalable, production-oriented AI applications for AI software engineering roles.

TECHNICAL SKILLS

LLM / GenAI: Retrieval-augmented generation, agentic AI, OpenAI APIs, MCP, embeddings, vector search, Hugging Face models, LLM tool integration
Machine Learning: PyTorch, TensorFlow, scikit-learn, reinforcement learning, quantization, neural networks, Optuna, Ray Tune, Hydra
Data & Visualization: Pandas, Matplotlib, Seaborn, Databricks, BigQuery, SQL Server, MongoDB
Programming: Python, JavaScript, TypeScript, React, Redux, Angular, Node.js, GraphQL, C#, .NET, REST APIs
Cloud, Infrastructure & Tooling: AWS, Google Cloud, Kubernetes, CI/CD, Jest, Cypress, Storybook, Material UI, Cursor

PROFESSIONAL EXPERIENCE

Senior Software Engineer II, **Nike**, Portland, Oregon — April 2022 to present

- Built a Figma automation pipeline that reduced content creation time by 80%, improving delivery speed and operational efficiency
- Developed a JavaScript MCP client that integrated LLM tools, internal APIs, and Databricks to enable AI-assisted workflows
- Designed and implemented an AI analytics innovation project, training AI models to analyze production data
- Led the server-driven storytelling project to revamp the Nike app
- Mentored team members
- Technologies: React, JavaScript, GraphQL, TypeScript, Node.js, Java, APIs, AWS, Kubernetes, Cursor

Full Stack Engineer, **Formative**, Remote — May 2021 to March 2022

- Built features for a startup education software with reusable UI components, improving user experience
- Technologies: React, Node.js, GraphQL, MongoDB

UI/UX Developer Lead, **KPA**, Portland, Oregon November 2019 to May 2021

- Led frontend development for a compliance platform, delivered features that generated $1.7M in operational savings, and drove modernization of a legacy application
- Technologies: Angular, Typescript, C#, .NET Core API, SQL

Senior C# Developer (Contract), **Daimler Trucks**, Portland, Oregon March 2018 to May 2019

- Re-architected enterprise application to improve performance, reliability, and platform stability
- Technologies: Angular, TypeScript, C#, .NET Core APIs, SQL

Full Stack Software Engineer, **OIA Global**, Portland, Oregon 2016 to 2017

- Implemented features for packaging logistics software: orders, shipments, and tracking
- Technologies: C#, SQL, JavaScript

Software Engineer, **Corvel**, Portland, Oregon 2014 to 2015

- Implemented functionality for health claims provider software
- Technologies: C#, Angular, JavaScript, Web API, SQL

Software Engineer, **Solera/GTS Services**, Portland, Oregon 2011 to 2013

- Implemented features for glass software application: scheduling, vendor sourcing, and workflows
- Technologies: C#, JavaScript, .NET, SQL

AI PROJECTS AND RESEARCH

Researcher, The University of Texas at Austin. Build optimization workflows for the Living and Working with Robots Lab using PyTorch, Python, neural networks, Optuna, Ray Tune, Hydra, visualizations, and TensorBoard.

Retrieval-Augmented Generation Chat System. Built an AI chat system to answer questions over clinical notes using semantic retrieval, document chunking, embeddings, vector search, and Hugging Face models.

Hospital Flow AI Agent. Built an agent-based AI system to predict ICU escalation risk and optimize patient flow using structured clinical data.

Health Indicator Prediction from Lifestyle and Biometric Data. Built an end-to-end pipeline to predict lifestyle-driven health outcomes. Trained recurrent neural network and tree-based classification models, created visualizations, and interpreted results. Published on EngrXiv.

EDUCATION

M.S. Artificial Intelligence, The University of Texas at Austin Expected December 2026

B.S. Computer Engineering, McGill University